Clinically-Aware Deep Learning for Osteoarthritis Severity Prediction from Hand X-rays

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Abstract

Accurate assessment of osteoarthritis (OA) severity is important for clinical diagnosis. The Kellgren–Lawrence (KL) grading system is widely used, but manual grading is subjective and inconsistent. In this project, we develop a deep learning framework to predict KL scores from hand X-ray images.

We address key challenges including class imbalance, ordinal label structure, and clinically asymmetric errors. Specifically, we explore ordinal regression, asymmetric loss design, and feature fusion with clinical attributes such as OP, JSN, and ER.

The proposed method aims to improve both prediction performance and clinical reliability.

1. Introduction

Osteoarthritis (OA) severity assessment is important for clinical diagnosis and treatment planning. The Kellgren–Lawrence (KL) grading system is commonly used to evaluate OA severity from X-ray images, with scores ranging from 0 to 4. However, KL grading often depends on manual interpretation, which may lead to inter-observer variability and limited scalability.

In this project, we focus on automatic KL score prediction from hand joint X-ray images using deep learning methods. Compared with standard image classification tasks, this problem has several challenges, including class imbalance, ordinal label structure, and clinically asymmetric errors. In particular, underestimation of OA severity may delay diagnosis and treatment.

To address these issues, we propose a clinically-aware deep learning framework. Starting from a CNN-based baseline, we explore imbalance handling, ordinal regression, asymmetric loss design, and feature fusion with clinical attributes such as OP, JSN, and ER. The goal is to improve both prediction performance and clinical reliability.

2. Related Work

Recent studies have shown that deep learning models can effectively predict osteoarthritis (OA) severity from X-ray images. Convolutional neural networks (CNNs) have been widely used for automatic Kellgren–Lawrence (KL) grading, demonstrating strong performance in extracting discriminative image features. However, prior work also highlights limitations in predicting intermediate KL grades and achieving consistent performance

across all severity levels (Tiulpin et al., 2018; Zhao et al., 2025).

A key limitation of many existing approaches is that KL grading is typically formulated as a standard multi-class classification problem. This ignores the ordinal nature of KL scores, where severity levels follow a natural progression. To address this, several studies have proposed ordinal regression methods, which improve prediction consistency by incorporating the ordered relationship between labels (Yong et al., 2022; Vente et al., 2021).

Class imbalance is another common challenge in OA datasets, where higher KL grades are underrepresented. Existing methods address this issue through resampling strategies and class-weighted loss functions to improve performance on minority classes (Hassan et al., 2024). In addition, recent work explores multimodal approaches that combine imaging data with clinical features, showing that structured attributes can provide complementary information for disease prediction (Rahim et al., 2023).

Despite these advances, most existing methods focus primarily on overall classification accuracy and do not explicitly consider the clinical impact of different types of prediction errors. In particular, underestimation of disease severity remains insufficiently addressed, even though it may lead to delayed diagnosis and treatment.

To address these limitations, our work integrates ordinal regression, imbalance-aware learning, and a clinically-aware asymmetric loss within a unified framework. In addition, we explore feature fusion with clinical attributes to improve both predictive performance and clinical reliability.

3. Dataset and Preprocessing

3.1 Dataset Description

The dataset used in this study consists of approximately 41,000 hand X-ray images of finger joints, including distal interphalangeal (DIP), proximal interphalangeal (PIP), and metacarpophalangeal (MCP) joints. Each image is annotated with Kellgren–Lawrence (KL) scores ranging from 0 to 4, representing the severity of osteoarthritis. In addition, auxiliary radiographic features are provided, including osteophytes (OP), joint space narrowing (JSN), and erosion (ER).

KL score distributions vary across different joint types. MCP joints exhibit severe class imbalance, with most samples concentrated in lower KL grades, while DIP joints show a more balanced distribution. Therefore, DIP joints are selected as the primary focus of this study to ensure more stable model training and evaluation.

3.2 Data Splitting

To avoid data leakage, the dataset is split at the patient level rather than the image level, ensuring that images from the same patient do not appear in multiple subsets. The data is divided into 70% training, 15% validation, and 15% testing sets, with stratification applied based on KL scores.

Table 1.

Distribution of KL scores across training, validation, and testing sets for DIP joints after balancing.

Set	KL =0	KL =1	KL =2	KL =3	KL =4	Total

Training	565	565	565	159	112	1966
Validation	130	130	130	17	26	433
Testing	160	160	160	39	32	551

3.3 Imbalance Handling

Despite stratified splitting, the dataset remains imbalanced, with higher KL grades underrepresented. To mitigate this issue, majority classes are downsampled to create a more balanced distribution across KL grades. This approach helps improve the model's sensitivity to higher severity cases, although it may reduce the total number of training samples.

3.4 Preprocessing and Augmentation

Image preprocessing is applied to improve data quality and model performance. Contrast Limited Adaptive Histogram Equalization (CLAHE) is used to enhance local contrast and highlight joint structures. All images are resized to 224×224, converted to three channels to match pretrained CNN inputs, and normalized.

To improve generalization, data augmentation techniques are applied during training, including random rotation (±10°), horizontal flipping, small translations, and brightness and contrast adjustments. These transformations introduce variability while preserving important anatomical features, helping reduce overfitting.

4. Methodology

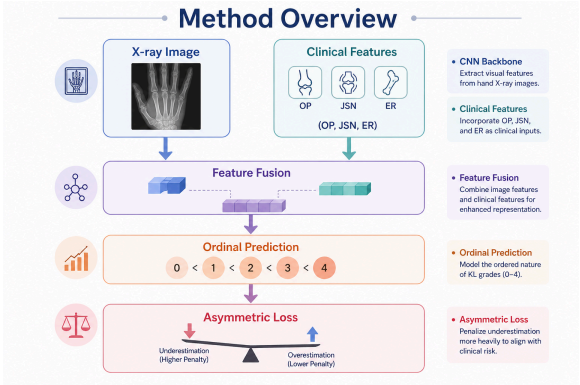
4.1 Overall Framework

The overall framework is illustrated in Figure 1. It integrates image-based feature extraction, clinical feature incorporation, and clinically-aware prediction objectives into a unified pipeline.

Given an input hand X-ray image, a convolutional neural network (CNN) backbone is first used to extract visual features. In parallel, structured clinical features, including osteophytes (OP), joint space narrowing (JSN), and erosion (ER), are incorporated as complementary inputs. These features are then combined through a feature fusion module. The model predicts KL grades as an ordinal outcome, reflecting the ordered nature of disease severity. To further align model optimization with clinical priorities, an asymmetric loss is applied, assigning higher cost to underestimation errors.

Figure 1.

Overview of the proposed framework. The model integrates image features and clinical features through a fusion module, followed by ordinal prediction and asymmetric loss optimization to align with clinical risk.



4.2 Model Design

We adopt ResNet50 with ImageNet-pretrained weights as the base architecture for KL grade prediction. The final fully connected layer is replaced with a task-specific classification head that outputs predictions for KL grades ranging from 0 to 4.

The model is fine-tuned on the target dataset using transfer learning.

To assess the impact of model capacity and architecture design, we additionally consider alternative backbone networks, including MobileNetV2 and DenseNet169. All models are trained under identical settings to ensure a consistent comparison.

4.3 Imbalance Handling

Class imbalance remains a challenge in KL grade prediction, particularly due to the underrepresentation of higher severity levels. While data-level balancing is applied during preprocessing, additional strategies are introduced at the model level to further mitigate this issue. Specifically, we adopt class-weighted loss to assign higher importance to minority classes during training. This approach helps reduce bias toward lower KL grades and encourages the model to better capture higher severity cases.

4.4 Ordinal Regression and Asymmetric Loss

KL score prediction is inherently an ordinal problem, where labels follow a natural progression from 0 to 4. Standard classification methods ignore this structure and treat all errors equally, regardless of their magnitude.

To address this limitation, we incorporate ordinal modeling to capture the relative distance between predicted and true labels, allowing the model to account for the severity of misclassification. However, not all prediction errors are equally important in clinical practice. In particular, underestimation of disease severity is more harmful than overestimation. To reflect this asymmetry, we introduce a clinically-aware asymmetric loss that assigns higher cost to underestimation errors.

Together, these two components provide a more comprehensive learning objective: ordinal modeling captures the magnitude of prediction errors, while asymmetric loss encodes their clinical significance.

4.5 Feature Fusion

In addition to image-based features, we incorporate structured clinical features, including osteophytes (OP), joint space narrowing (JSN), and erosion (ER), which are known to be associated with osteoarthritis severity. These features provide complementary information to visual representations extracted from X-ray images, capturing structural characteristics that may not be fully reflected in pixel-level patterns.

To integrate these heterogeneous inputs, image features and clinical features are combined through a feature fusion module to produce a unified representation for KL grade prediction.

4.6 Evaluation

Model performance is evaluated using a set of metrics designed to capture classification ability, ordinal consistency, and clinical relevance. Rather than relying on a single metric, we select complementary measures that align with the key objectives of this task, including disease screening, severity grading, and reduction of clinically harmful errors.

From a screening perspective, binary AUC is used to evaluate the model's ability to distinguish disease (KL 3–4) from non-disease (KL 0–2), based on predicted probabilities.

To assess grading performance under clinical constraints, we adopt a clinically-aware metric, ω-MAE, which extends mean absolute error by assigning higher penalties to underestimation. It is defined as:

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where W(·) is an asymmetric cost matrix that penalizes underestimation more heavily than overestimation, aligning evaluation with clinical risk considerations (Aktemur & Öksüz, 2023).

In addition, the underestimation rate is reported to directly quantify the proportion of predictions where severity is underestimated. Per-class recall is used to evaluate performance across KL grades, with particular emphasis on higher severity levels, while overall accuracy is included as a supplementary reference.

The evaluation metrics used in this study are summarized in Table 2.

Table 2.

Summary of evaluation metrics for KL grading, including classification, ordinal, and clinically-informed measures.

Metric	Question Addressed	Notes
AUC / ROC	How well does the model distinguish disease (KL 3–4) from non-disease (KL 0–2)?	Based on probability scores
ω-MAE	What is the clinically-weighted prediction error?	Penalizes underestimation more heavily
Underestimation Rate	How often does the model underestimate severity?	Direct measure of clinical risk
Per-class Recall	How well are individual KL grades identified?	Focus on KL 3–4
Overall Accuracy	What proportion of predictions are correct?	Supplementary metric

5 Experiment

5.1 Loss Ablation

This experiment aims to validate the effectiveness of the proposed asymmetric loss design. We compare standard cross-entropy (CE), class-weighted CE, and the proposed asymmetric loss on ResNet50. Class weighting is included as a commonly used strategy for handling class imbalance, which has been shown to provide relatively stable improvements across different backbone architectures, while more complex loss formulations such as focal loss may exhibit inconsistent performance (Hoang et al., 2023) . The asymmetric loss is designed based on clinical considerations, where underestimation of disease severity is more harmful than overestimation and is therefore penalized more heavily.

As shown in Table 3, the asymmetric loss without class weighting achieves the best performance across all primary metrics. The underestimation rate is reduced from 0.208 to 0.101 (−51%), and ω-MAE decreases from 0.030 to 0.025 (−16%), confirming that explicitly penalizing underestimation improves clinically relevant performance. However, this improvement comes with a trade-off: KL0 recall becomes unstable, with some runs collapsing to zero, indicating an overcompensation effect toward higher severity predictions. This suggests that while asymmetric loss aligns model behavior with clinical priorities, loss design alone is insufficient to ensure balanced performance, motivating the need for improved feature learning in subsequent experiments. To further illustrate this effect, the confusion matrices for the baseline and asymmetric loss models are shown in Figures 2 and 3.

Table 3.

Loss ablation results on ResNet50, comparing classification, ordinal, and clinically-informed metrics.

Model	Binary AUC	ω-MAE	Underestimation Rate

CE (no weight)	0.9902	0.0295	0.2081
CE (weight)	0.9853	0.0294	0.2003
Asym (no weight)	**0.9918**	**0.0247**	**0.1010**
Asym (weight)	0.9908	0.0278	0.1016

Figure 2.
Confusion matrix for the baseline model trained with cross-entropy loss.

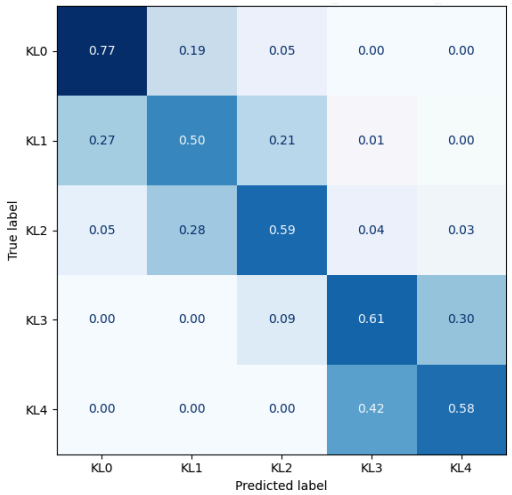
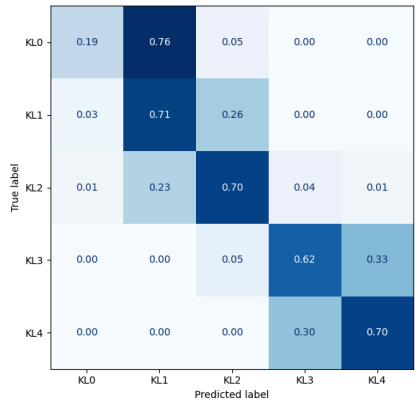


Figure 3.
Confusion matrix for the model trained with asymmetric loss (no class weighting), showing reduced underestimation errors compared to the baseline.



5.2 Backbone Selection

This experiment aims to identify an appropriate backbone architecture under the fixed asymmetric loss setting. We compare several widely used CNN architectures, including ResNet, DenseNet, and MobileNet variants, which have been commonly adopted in knee osteoarthritis prediction tasks and shown to exhibit varying performance across models (Va et al., 2023).

ResNet50 and ResNet101 exhibit similar performance and both show unstable behavior in KL0 recall, with collapse observed in some runs, indicating that deeper residual structures do not alleviate the instability introduced by asymmetric loss (Table 4). DenseNet201 achieves the lowest ω-MAE, suggesting strong ordinal consistency, but shows a higher underestimation rate (0.117), which increases clinical risk. In contrast, MobileNetV2 achieves the highest Binary AUC (0.9959) and maintains a low underestimation rate (0.0926), providing the best balance between disease detection and risk control. Considering that the primary clinical objective is to maximize detection of severe cases while minimizing underestimation, MobileNetV2 is selected as the backbone for subsequent experiments. In addition, its

lightweight architecture allows more flexible model extensions in later stages. The confusion matrix of the selected MobileNetV2 model is shown in Figure 4, illustrating its relatively balanced prediction distribution across KL grades and stable performance on higher severity levels.

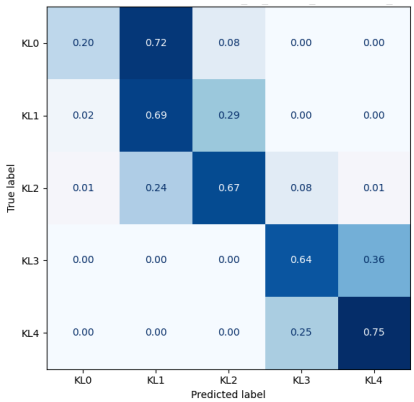
Table 4.

Backbone comparison using asymmetric loss (no class weighting).

Model	Binary AUC	ω-MAE	Underestimation Rate
ResNet50 (baseline)	0.9918	0.0247	0.1010
ResNet101	0.9906	0.0251	0.0847
DenseNet201	0.9922	**0.0223**	0.1174
MobileNetV2	**0.9959**	0.0251	**0.0926**

Figure 4.

Confusion matrix of the MobileNetV2 model with asymmetric loss (no class weighting), showing balanced prediction performance across KL grades.



5.3 Ordinal Methods

To explore whether explicitly modeling ordinal relationships can improve KL grade prediction, we build upon the baseline model from Experiment 5.2, which uses a MobileNetV2 backbone with asymmetric loss. Given that KL grading (0–4) is inherently ordinal, we investigate three representative ordinal regression methods, including ORM, CORAL, and CORN.

ORM formulates the task by learning ordered decision boundaries between adjacent grades (Yong et al., 2022). CORAL enforces rank consistency through shared classifier weights and monotonic constraints (Cao et al., 2020), while CORN models conditional dependencies between ordinal levels in a sequential manner (Shi et al., 2023). These methods aim to better capture the inherent ordering structure of KL grades.

However, experimental results show that none of these methods outperform the baseline. Although all approaches achieve similarly high binary AUC (≈0.99), ordinal performance degrades, with higher ω-MAE and underestimation rates. In particular, CORAL shows severe instability, and CORN, while more stable, still underperforms compared to the baseline.

These results suggest that strict ordinal modeling is not robust under class imbalance and limited data. Instead, directly optimizing asymmetric objectives provides more stable and clinically aligned performance. Based on these findings, we select the MobileNetV2 model with asymmetric loss as the baseline for subsequent experiments.

Table 5.

Ordinal methods comparison using asymmetric loss (no class weighting).

Model	Binary AUC	ω-MAE	Underestimation Rate
MobileNetV2+ (Asymmetric Loss)	0.9959	**0.0251**	**0.0926**
ORM	0.9955	0.0259	0.1331
CORAL	0.9936	0.0764	0.3811
CORN	**0.9960**	0.0315	0.2480

5.4 MSFM-Based Improvement

To further improve KL grade prediction, we explore whether enhancing feature representation can benefit model performance. Specifically, we introduce a Multi-Scale Feature Map fusion module (MSFM) to capture complementary information across different feature levels within the MobileNetV2 backbone (Rajpoot et al., 2025). We evaluate MSFM under different learning strategies, including cross-entropy, asymmetric loss, and ordinal-based designs.

Experimental results show that MSFM alone does not improve performance and may introduce instability. When combined with cross-entropy or ordinal methods, the model exhibits higher ω-MAE and underestimation rates, indicating that feature fusion without proper optimization can degrade performance.

However, when combined with asymmetric loss, MSFM achieves the best overall performance, with the lowest ω-MAE (0.0222) and underestimation rate (0.0914) while maintaining high AUC. These results suggest that effective feature enhancement depends on appropriate loss design. Therefore, we select the MobileNetV2 model with MSFM and ordinal asymmetric loss as the final model configuration. The confusion matrix of the final model is shown in Figure 5, illustrating its overall prediction behavior under the combined MSFM and asymmetric loss design.

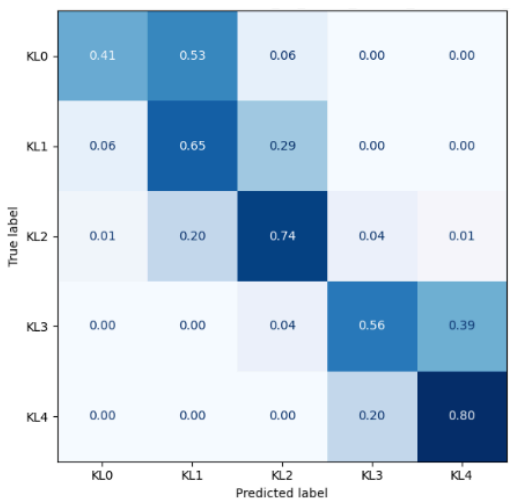
Table 6.

MSFM-based model comparison (no class weighting).

Model	Binary AUC	ω-MAE	Underestimation Rate
MobileNetV2+ (Asymmetric Loss)	**0.9959**	0.0251	0.0926
mobilenet_v2_MSFM (crossentropy)	0.9951	0.0253	0.1845
mobilenet_v2_MSFM (ordinal_asymmetric)	0.9955	**0.0222**	**0.0914**
mobilenet_v2_MSFM_CO	**0.9959**	0.0282	0.2299

RN			
mobilenet_v2_MSFM_ordinal	0.9958	0.0363	0.2656

Figure 5.
Confusion matrix of the final model (MobileNetV2 with MSFM and asymmetric loss), illustrating prediction behavior across KL grades.



5.5 Feature Fusion with CrossAttention

To explore the upper bound of model performance, we investigate whether incorporating additional clinical structural features can further improve KL grade prediction. Specifically, we extend the model from Section 5.4 by integrating auxiliary features (JSN, OP, ER) using multitask learning and cross-attention-based fusion (Wang et al., 2025; Lakshmi & Pushpa, 2026).

Experimental results show that incorporating structural features does not improve overall performance. Single-feature models significantly degrade performance, while multi-feature fusion improves severe-case detection (KL3/KL4 recall) but increases ω-MAE and underestimation. Cross-attention further enhances focus on severe cases, achieving the best KL3 recall and macro AUC, but leads to worse overall accuracy and higher prediction error.

These results indicate that additional feature fusion introduces complexity that exceeds the capacity of the current dataset, leading to unstable optimization. As a result, it does not provide consistent improvement over the baseline. Therefore, the best-performing model remains the MobileNetV2-based architecture enhanced with MSFM and optimized using ordinal asymmetric loss.

Table 7.
Feature Fusion comparison (no class weighting).

Model	Binary AUC	ω-MAE	Underestimation Rate
mobilenet_v2_MSFM (ordinal_asymmetric)	**0.9955**	**0.0222**	0.0914
MSFM_jsn	0.9954	0.0365	0.1077
MSFM_op	0.9863	0.0456	0.1083
MSFM_jsn_op	0.9939	0.0295	0.1446
MSFM_jsn_op_er	0.9943	0.0351	0.1022

MSFM_crossattn	0.9943	0.0410	**0.0538**

5.6 Summary of Results

To summarize the effectiveness of different components, we present the performance of key models across different stages of the framework. Table 8 shows the progression from the baseline model to the final model.

Table 8.

Overall Performance Comparison Across Model Variants

Stage	Model	Binary AUC	ω-MAE	Underestimation Rate
Baseline	ResNet50 + CE (no weight)	0.9902	0.0295	0.2081
Best Loss	ResNet50 + Asymmetric (no weight)	0.9918	0.0247	0.1010
Best Backbone	MobileNetV2 + Asymmetric	0.9959	0.0251	0.0926
Final Model	MSFM + Asymmetric	0.9955	0.022	0.0914

The results demonstrate a clear progression in model performance across different stages of the framework. Compared with the baseline (ResNet50 + CE), the final model (MobileNetV2 + MSFM + asymmetric loss) achieves substantial improvement in clinically relevant metrics. In particular, the underestimation rate is reduced from 0.208 to 0.091 (−56%), and ω-MAE decreases from 0.0295 to 0.0222, indicating more accurate and clinically safer predictions. Binary AUC is also consistently improved, reflecting strong disease detection capability.

More importantly, the results highlight that performance gains are primarily driven by clinically-aware loss design and improved feature representation. In contrast, explicit ordinal modeling and naive feature fusion do not provide consistent benefits under the current data setting, often leading to instability. These findings suggest that directly optimizing asymmetric objectives, combined with effective feature learning, is more robust for KL severity prediction than increasing model complexity.

Overall, the final model achieves the best balance between detection performance, ordinal consistency, and clinical reliability.

6 Result

This section interprets the experimental findings from a clinical perspective and highlights the key insights of the proposed framework.

The most significant improvement comes from the introduction of the asymmetric loss. Compared with the baseline

model, the underestimation rate is reduced from 0.2081 to 0.1010, indicating a substantial reduction in clinically harmful prediction errors. This confirms that explicitly penalizing underestimation is critical for aligning model behavior with real-world diagnostic priorities.

The backbone comparison further shows that model selection should consider both detection performance and clinical risk. While DenseNet201 achieves lower ω-MAE, its higher underestimation rate makes it less suitable in clinical settings. MobileNetV2, in contrast, provides the best trade-off, achieving high AUC while maintaining a relatively low underestimation rate.

The results of ordinal regression methods (ORM, CORAL, CORN) suggest that explicitly modeling ordinal structure does not consistently improve performance under class imbalance. In several cases, these methods increase underestimation or introduce instability, indicating that strict ordinal constraints may not be robust for this task.

The incorporation of multi-scale feature modeling (MSFM) further improves performance when combined with asymmetric loss, achieving the best overall results. This suggests that effective feature representation and clinically-aware loss design are complementary and should be jointly optimized.

In contrast, feature fusion with additional clinical attributes does not lead to consistent improvement. The increased model complexity introduces instability and does not translate into better clinical performance under the current data setting.

Overall, the final model achieves a strong balance between disease detection, ordinal prediction accuracy, and clinical safety. Most importantly, it substantially reduces underestimation risk, addressing a key limitation of existing approaches and improving the model's suitability for real-world clinical application.

7 Discussion

The experimental results provide several important insights for clinically-aware KL prediction. First, the effectiveness of the asymmetric loss highlights the importance of aligning optimization objectives with clinical priorities. By explicitly penalizing underestimation, the model reduces clinically harmful errors, suggesting that objective design plays a more critical role than increasing model complexity. This is further supported by the observation that standard ordinal regression methods do not consistently improve performance under class imbalance, whereas directly optimizing asymmetric objectives leads to more stable and clinically meaningful results.

Second, the results indicate that feature representation and loss design should be jointly considered. While MSFM alone does not significantly improve performance, its combination with asymmetric loss leads to the best overall results. This suggests that improving feature quality is most effective when guided by clinically-aware optimization objectives.

Despite these improvements, several limitations remain. The asymmetric loss introduces a bias toward higher KL predictions, leading to instability in KL0 recall and occasional class collapse. This

reflects a fundamental trade-off between reducing underestimation and preserving sensitivity to low-severity cases. In addition, the limited dataset size restricts the effectiveness of more complex modeling strategies, such as feature fusion and cross-attention, which require larger and more balanced data to achieve stable improvements.

Future work can address these limitations in two directions. First, more balanced or larger-scale datasets may help stabilize predictions across all KL grades and improve the effectiveness of feature fusion methods. Second, adaptive or hybrid loss designs could be explored to better balance underestimation penalties with low-grade classification performance, reducing the risk of overcompensation.

Overall, this study suggests that clinically-aware objective design, combined with effective feature learning, is a promising direction for improving both prediction performance and reliability in real-world medical applications.

8 Conclusion

In this work, we propose a clinically-aware deep learning framework for automatic KL score prediction from hand X-ray images. By incorporating asymmetric loss, backbone optimization, and multi-scale feature modeling, the proposed approach systematically improves both prediction performance and clinical reliability.

The main contribution of this study is to demonstrate that clinically-aware objective design plays a more critical role than increasing model complexity. In particular, asymmetric loss substantially reduces underestimation, leading to safer and more reliable predictions in real-world settings. Furthermore, the combination of asymmetric loss and multi-scale feature modeling achieves the best overall performance, highlighting the importance of jointly optimizing loss design and feature representation.

Beyond performance improvement, our results provide important insights into model design under limited data conditions. Specifically, we show that strict ordinal modeling and complex feature fusion strategies do not consistently improve performance, and may introduce instability. In contrast, directly optimizing for clinical risk leads to more robust and interpretable behavior.

Future work may focus on improving data availability and developing adaptive loss functions to better balance underestimation control and low-grade sensitivity. In addition, more effective multimodal fusion strategies could be explored under larger-scale datasets to further enhance performance.

Overall, this work highlights the importance of aligning deep learning models with clinical priorities and provides a practical approach for improving both accuracy and reliability in medical image analysis.

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Appendix A. Detailed Experimental Results

Primary evaluation metrics (Binary AUC, ω-MAE, and Under Rate) are reported in the main text. This appendix provides additional results, including overall accuracy and per-class recall (KL0–4), for completeness.

Table B1.
Detailed per-class performance for Loss Ablation experiments (mean ± std over 3 runs).

Model	Accuracy	KL0	KL1	KL2	KL3	KL4
CE (no weight)	0.6177 ± 0.0008	0.7688 ± 0.0184	0.5000 ± 0.0357	0.5937 ± 0.0399	0.6068 ± 0.0735	0.5833 ± 0.0531
CE (weight)	0.6243 ± 0.0102	0.7354 ± 0.0358	0.5729 ± 0.0281	0.5833 ± 0.0574	0.5555 ± 0.0604	0.6146 ± 0.0147
Asym (no weight)	0.5499 ± 0.0745	0.1938 ± 0.2740	0.7083 ± 0.0524	0.7021 ± 0.0511	0.6154 ± 0.0209	0.6979 ± 0.0736
Asym (weight)	0.4991 ± 0.0541	0.1771 ± 0.2504	0.6625 ± 0.0802	0.6042 ± 0.0397	0.5385 ± 0.0209	0.7188 ± 0.0255

The asymmetric loss introduces substantial variability in KL0 recall, indicating instability in low-severity predictions.

Table A2.
Detailed per-class performance for backbone comparison under asymmetric loss (mean ± std over 3 runs).

Model	**Accuracy**	**KL0**	**KL1**	**KL2**	**KL3**	**KL4**
ResNet101	0.5293 ± 0.0575	0.1687 ± 0.2386	0.6333 ± 0.0780	0.7250 ± 0.0265	0.6325 ± 0.0436	0.7083 ± 0.0589
DenseNet201	**0.6370 ± 0.0051**	**0.6041 ± 0.0587**	0.5979 ± 0.0213	**0.7146 ± 0.0369**	0.5983 ± 0.0436	0.6562 ± 0.0920
MobileNetV2	0.5414 ± 0.0766	0.1979 ± 0.2799	**0.6917 ± 0.0428**	0.6688 ± 0.0270	**0.6410 ± 0.0363**	**0.7500 ± 0.0510**

MobileNetV2 achieves stronger performance on higher severity levels (KL3–KL4), while DenseNet201 shows more stable predictions for lower severity levels (KL0). However, MobileNetV2 maintains a better balance between high-severity detection and underestimation control, supporting its selection in the main text.

Table A3.
Detailed per-class performance for ordinal methods under asymmetric baseline (mean ± std over 3 runs).

Model	Accuracy	KL0	KL1	KL2	KL3	KL4
ORM	**0.6243 ± 0.0039**	0.7375 ± 0.0385	0.3458 ± 0.0230	**0.7813 ± 0.0664**	0.4273 ± 0.0673	**0.9063 ± 0.0442**
CORAL	0.4096 ± 0.0048	**0.9417 ± 0.0078**	**0.0000 ± 0.0000**	0.2687 ± 0.0102	**0.0000 ± 0.0000**	**1.0000 ± 0.0000**
CORN	0.6038 ± 0.0141	0.8042 ± 0.0030	0.4708 ± 0.0358	0.5312 ± 0.0518	**0.5641 ± 0.0554**	0.6771 ± 0.0531

CORAL exhibits severe prediction collapse, with zero recall for intermediate grades (KL1 and KL3), while CORN shows improved stability but significantly higher underestimation risk. ORM achieves reasonable overall accuracy but performs poorly on intermediate severity levels. These results highlight the instability of standard ordinal methods in this setting.

Table A4.
Detailed per-class performance for MSFM-based models under different learning strategies (mean ± std over 3 runs).

Model	Accuracy	KL0	KL1	KL2	KL3	KL4
MSFM + CE	**0.6497 ± 0.0015**	**0.7562 ± 0.0234**	0.5333 ± 0.0128	0.6750 ± 0.0270	0.5983 ± 0.0242	0.6354 ± 0.0390
MSFM + Asym	0.6092 ± 0.0717	0.4062 ± 0.2892	**0.6500 ± 0.0503**	**0.7438 ± 0.0204**	0.5641 ± 0.2019	**0.8021 ± 0.1405**
MSFM + CORN	0.6310 ± 0.0067	0.7958 ± 0.0107	0.5312 ± 0.0222	0.5771 ± 0.0369	**0.6239 ± 0.0639**	0.5833 ± 0.0390

MSFM + Ordinal Head	0.5414 ± 0.0074	**0.8896 ± 0.0059**	0.2771 ± 0.0078	0.4854 ± 0.0252	**0.0513 ± 0.0209**	**1.0000 ± 0.0000**

MSFM combined with asymmetric loss achieves strong performance on higher severity levels (KL2–KL4), while maintaining improved overall balance compared to other configurations. In contrast, MSFM with cross-entropy shows more stable low-severity predictions but weaker performance on higher KL grades.

The introduction of ordinal structures leads to unstable behavior, particularly in the ordinal head setting, where severe collapse is observed (e.g., KL3 recall ≈ 0 and KL4 recall = 1.0), further confirming the limitations of ordinal modeling under limited data.

These results support the selection of MSFM with asymmetric loss as the final model configuration.

Table A5.
Detailed per-class performance for multitask feature fusion models (mean ± std over 3 runs).

Model	Accuracy	KL0	KL1	KL2	KL3	KL4
MSFM + JSN	0.4168 ± 0.0400	**0.0000 ± 0.0000**	0.4896 ± 0.3464	0.6688 ± 0.1859	0.6581 ± 0.2097	0.5833 ± 0.4125
MSFM + OP	0.3545 ± 0.0449	**0.0000 ± 0.0000**	0.2187 ± 0.3093	0.8104 ± 0.1593	0.7863 ± 0.0121	**0.0000 ± 0.0000**
MSFM + JSN + OP	0.4652 ± 0.0098	**0.0000 ± 0.0000**	**0.7188 ± 0.0405**	0.6542 ± 0.0059	**0.9402 ± 0.0121**	**0.0000 ± 0.0000**
MSFM + Multitask	0.4350 ± 0.0665	**0.0000 ± 0.0000**	0.4917 ± 0.3486	0.7458 ± 0.1401	0.8461 ± 0.1451	0.2708 ± 0.3830
MSFM + CrossAttn	0.4041 ± 0.0677	**0.0000 ± 0.0000**	0.2375 ± 0.3359	**0.8521 ± 0.1343**	0.7949 ± 0.1308	0.5417 ± 0.3830

All multitask feature fusion variants exhibit systematic collapse in KL0 prediction, with KL0 recall consistently equal to 0 across all runs. This indicates that the introduction of auxiliary tasks (JSN, OP, ER) leads to strong optimization conflicts with the main KL prediction task, particularly for low-severity samples.

While some models achieve high recall for specific higher severity levels (e.g., KL3 in MSFM + JSN + OP, KL2 in CrossAttn), this comes at the cost of severely degraded overall

prediction balance. The instability is further reflected in large variances across runs for multiple classes.

These results suggest that, under limited data conditions, increasing model complexity through multitask learning introduces more harm than benefit. The model tends to prioritize high-severity predictions while completely sacrificing low-severity detection, making it unsuitable for clinical use.

Appendix B. Additional Confusion Matrices

In addition to the quantitative metrics reported in the main text, this appendix presents confusion matrices for different model variants to provide a more intuitive understanding of prediction behavior across KL grades.

These visualizations complement the experimental results by illustrating how different modeling strategies affect class-wise predictions. In particular, they highlight several important patterns observed in this study. First, baseline models tend to exhibit underestimation bias, with predictions skewed toward lower KL grades. Second, asymmetric loss reduces underestimation but introduces instability in low-severity predictions, occasionally leading to KL0 collapse. Third, ordinal regression methods show inconsistent behavior, with some models failing to correctly distinguish intermediate grades. Finally, more complex models, such as multitask learning and cross-attention, often lead to unstable pred

iction distributions, including severe class imbalance and collapse in specific KL categories.

Overall, these confusion matrices provide visual evidence supporting the conclusions in the main text, demonstrating that clinically-aware loss design and appropriate feature representation are more effective than increasing model complexity under limited data conditions.

Figure B1.

Confusion matrix for the ResNet50 model trained with cross-entropy loss with class weighting.

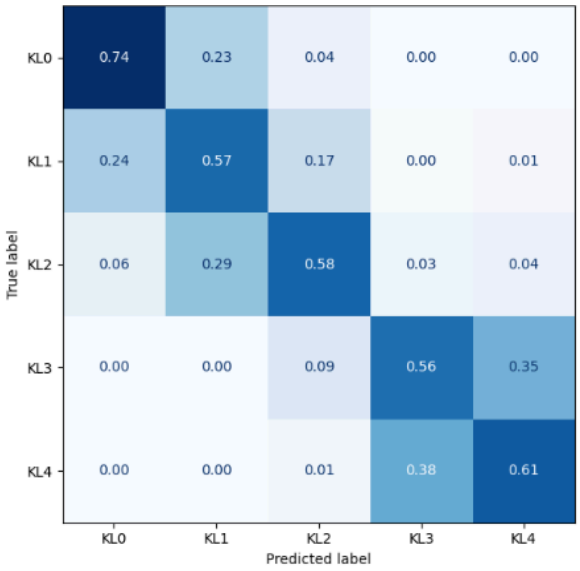


Figure B2.

Confusion matrix for the ResNet50 model trained with ordinal asymmetric loss using class weighting.

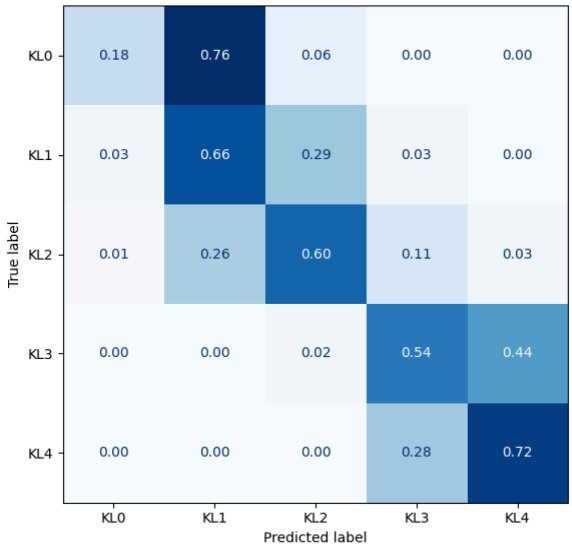


Figure B3.

Confusion matrix for the ResNet101 model with asymmetric loss (no class weighting)

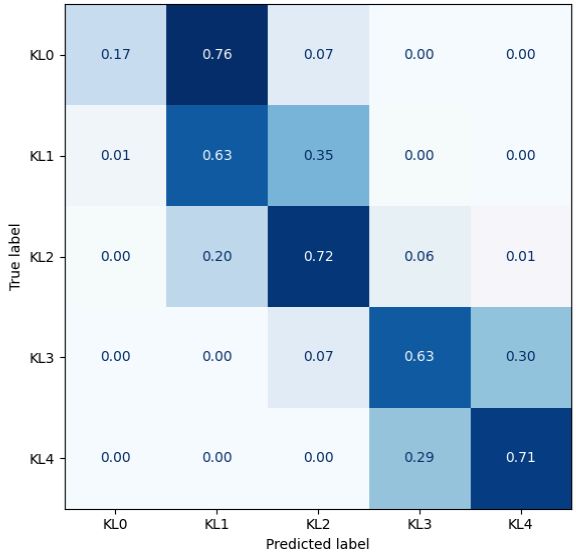


Figure B4.

Confusion matrix for the DenseNe201 model with asymmetric loss (no class weighting)

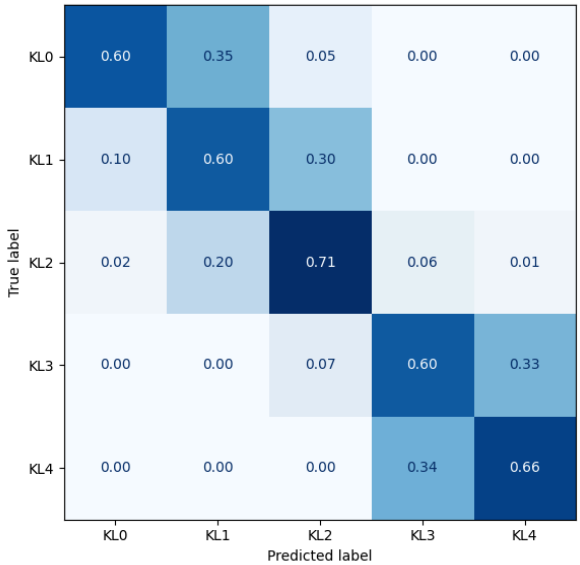


Figure B5.
Confusion matrix for the MobileNetV2 model with ORM with asymmetric loss (no class weighting).

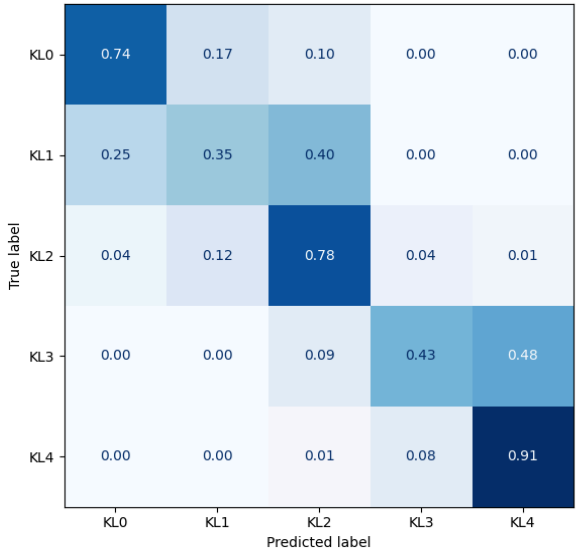


Figure B6.
Confusion matrix for the MobileNetV2 model with CORAL with asymmetric loss (no class weighting).

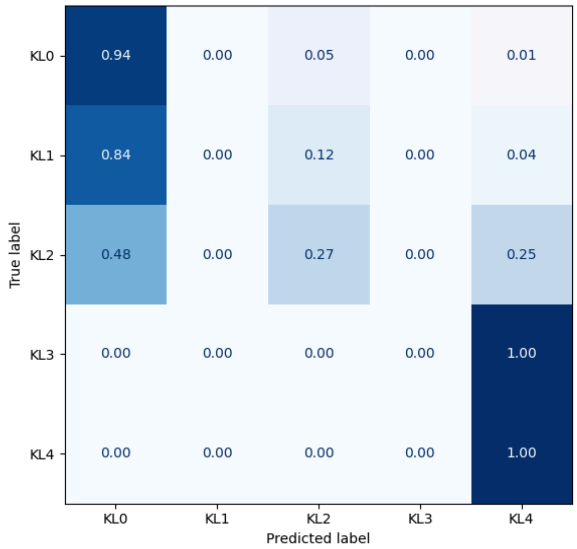


Figure B7.
Confusion matrix for the MobileNetV2 model with CORN with asymmetric loss (no class weighting).

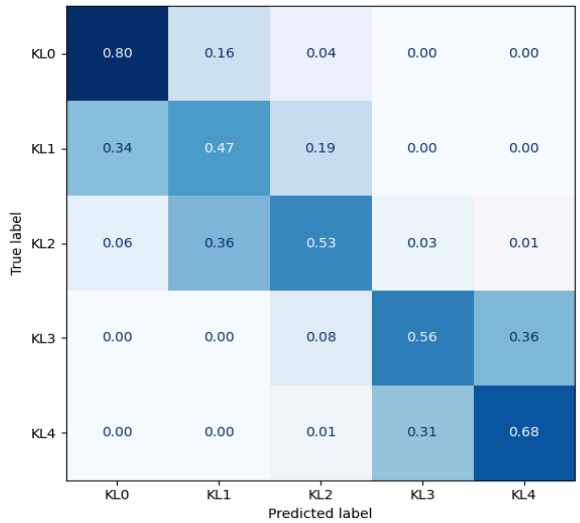


Figure B8.
Confusion matrix for the MobileNetV2 model with MSFM and cross-entropy loss (no class weighting).

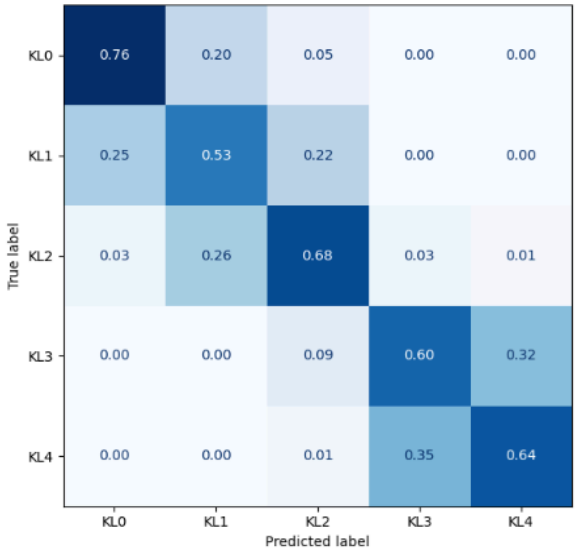


Figure B9.

Confusion matrix for the MobileNetV2 model with MSFM and ordinal (no class weighting).

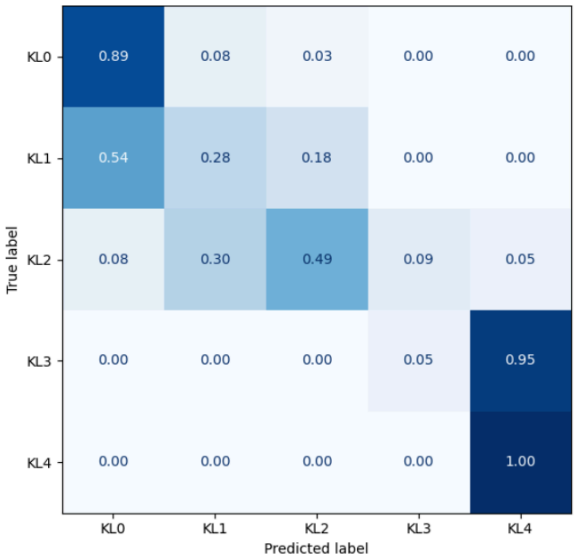


Figure B10.

Confusion matrix for the MobileNetV2 model with MSFM and CORN loss (no class weighting).

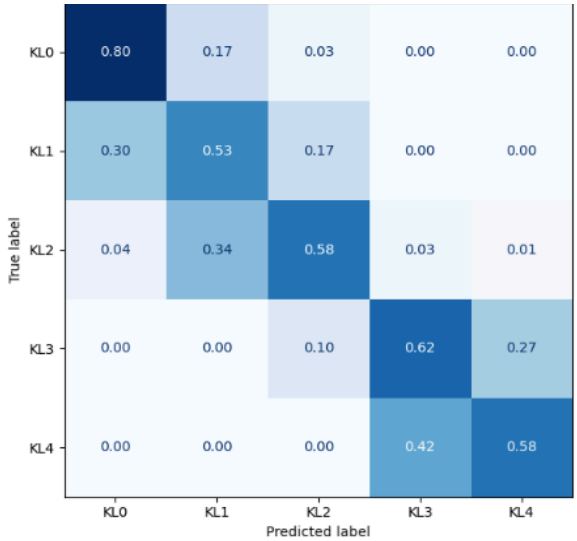


Figure B11.
Confusion matrix for the MobileNetV2 model with MSFM and JSN multitask + asymmetric loss (no class weighting).

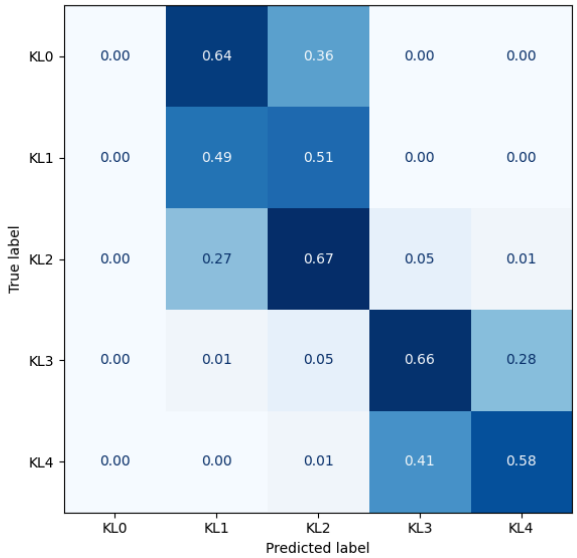


Figure B12.
Confusion matrix for the MobileNetV2 model with MSFM and OP multitask + asymmetric loss (no class weighting).

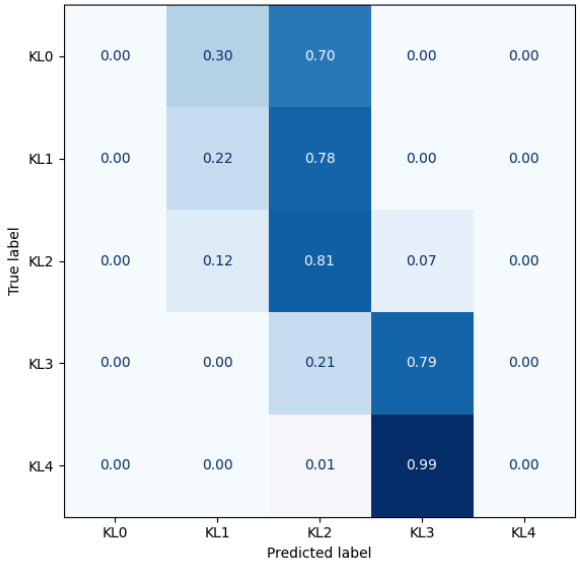


Figure B13.
Confusion matrix for the MobileNetV2 model with MSFM and JSN&OP multitask + asymmetric loss (no class weighting).

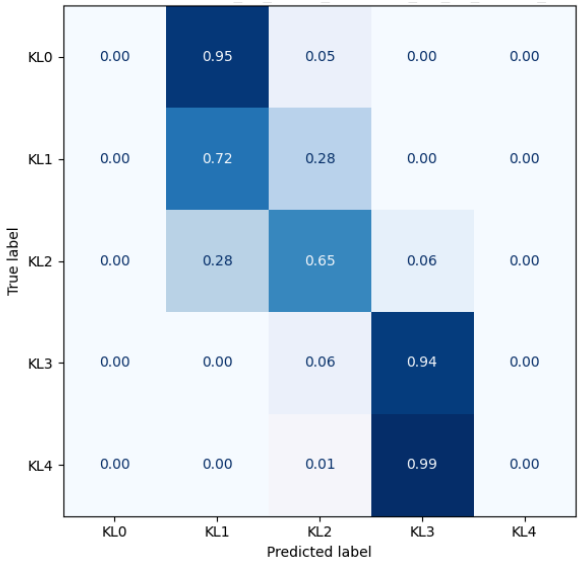


Figure B14.
Confusion matrix for the MobileNetV2 model with MSFM and JSN&OP&ER multitask + asymmetric loss (no class weighting).

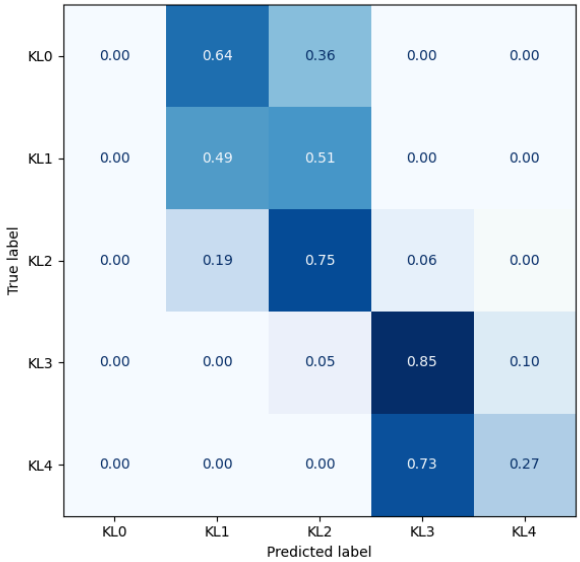
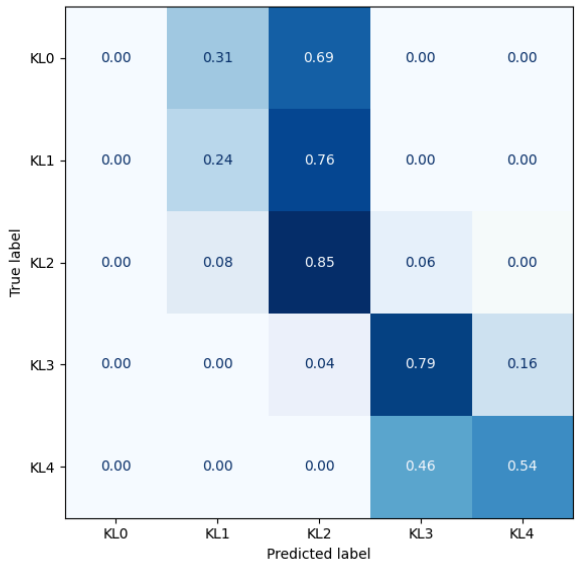


Figure B15.

Confusion matrix for the MobileNetV2 model with MSFM and JSN&OP&ER multitask + crossattention and asymmetric loss (no class weighting).